Syllabus for Electromagnetic Theory: PHYS330

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Abstract

This course provides a rigorous exploration of classical electrodynamics, including new computational electromagnetism (CEM) content. The course begins with a review of vector calculus and its physical interpretation in electromagnetism. Next, the electrostatics of charge distributions using Coulomb's law, Gauss's law, and the electric potential will be presented. The course proceeds to boundary-value problems and solutions to Laplace's equation, including multipole expansions and the method of images. Building on the introductory content, the course procedes to properties and behavior of electric fields in matter, covering polarization, bound charges, and the electric displacement field. From electrostatics, the course proceeds to magnetostatics, introducing the Biot–Savart law, Ampère's law, vector potential, and the role of magnetization in materials. New CEM content serves as an introduction to time-dependent electromagnetic fields. Throughout the course, emphasis is placed on analytical problem-solving through active learning strategies. Final projects allow students to creatively explore scientific and engineering applications of electrodynamics and CEM.

Pre-requisites*: PHYS150, PHYS180, PHYS185, PHYS275, MATH141, MATH142, MATH241*
Course credits, Liberal Arts Categorization*: 3.0 Credits, None*
Regular course hours*: TR 11:00am-12:20pm, SLC 232.*
Instructor contact information*: Discord: 918particle, Email: jhanson2@whittier.edu, Office: SLC 222.*
Office hours*: Open-door policy, or make an appointment via Discord/email.*
Attendance/Absence*: Students needing to reschedule midterms must notify the professor a few days in advance.*
Late work policy*: Late work is generally not accepted, but is left to the discretion of the instructor.*
Text: *Introduction to Electrodynamics, 5th ed. by D. Griffiths -* `https://doi.org/10.1017/9781009397735`.
Grading*: In-class warm-up exercises are marked only for attendance. Weekly homework sets are intended to be challenging, and student collaboration is expected. Two short, open-book midterm exams will be completed outside of class. There is no final exam for the course. A final project and class presentation will be completed near the end of the semester. See Tab. 1.*
Grade Settings*: $< 60\%$ = F, $\geq 60\%, < 70\%$ = D, $\geq 70\%, < 80\%$ = C, $\geq 80\%, < 90\%$ = B, $\geq 90\%, < 100\%$ = A. Pluses and minuses: 0-3% minus, 3%-6% straight, 6%-10% plus (e.g. 79% = C+, 91% = A-)*
Final projects*: Students may contribute one of two types of final project: A) a more lengthy or challenging problem to teach the class, or B) a computational electromagnetism (CEM) demonstration. The final presentation should be approximately 20 minutes. Students should begin coding and producing basic solutions to challenge problems early in the course to facilitate creation of the final presentation.*
ADA Statement on Disability Services*: Whittier College is committed to make learning experiences as accessible as possible. If you experience physical or academic barriers due to a disability, you are encouraged to contact Student Disability Services (SDS) to discuss options. To learn more about academic accommodations, email disabilityservices@whittier.edu, call (562) 907-4825, or go to SDS which is located on the ground floor of Wardman Library.*
Academic Honesty Policy*:* `http://www.whittier.edu/academics/academichonesty`
Course Objectives*:*

- To master the application of physical units, taking limits, and symmetry in physics
- To master techniques in mathematical physics and CEM and apply them to theoretical physics and engineering
- To use active learning techniques like directed warm-up exercises, peer learning, simulations, and student-designed final projects to build a better understanding of electrodynamics

Assignment	**Weight**	**Assigned**	**Due**
Problem Sets	40%	See outline (next page)	Two weeks after assigned
Midterm 1	15%	Friday, October 10th, 2025	Monday, October 13th, 2025
Midterm 2	15%	Friday, November 14th, 2025	Monday, November 17th, 2025
Final Project Proposal	10%	Friday, November 21st, 2025	Friday, November 24th, 2025
Final Project Presentation	20%	Friday, November 21st, 2025	Thursday, December 4th, 2025

Table 1: Assignments for Electromagnetic Theory (PHYS330).

Course Outline:

1. **Vector Analysis, Chapter 1**
   (a) Vector algebra
   (b) Differential calculus in three dimensions
   (c) Integral calculus in three dimensions
   (d) Cylindrical and spherical coordinate systems
   (e) The Dirac Delta function
   (f) Theory of vector fields
   (g) *Homework 1, assigned August 22nd*
2. **Electrostatics, Chapter 2**
   (a) The electric field
   (b) Divergence and curl of electrostatic fields
   (c) Electric potential
   (d) Work and energy in electrostatics
   (e) Conductors
   (f) *Homework 1, due September 5th*
   (g) *Homework 2, assigned September 5th*
3. **Potentials, Chapter 3**
   (a) Laplace's equation
   (b) The method of images
   (c) Separation of variables
   (d) Multipole expansion
   (e) *Homework 2, due September 19th*
   (f) *Homework 3, assigned September 19th*
4. **Electric Fields in Matter, Chapter 4**
   (a) Polarization
   (b) The field of a polarized object
   (c) The electric displacement
   (d) Linear dielectrics
   (e) *Homework 3, due October 3rd*
5. **Midterm 1, Chapters 1-4**
   (a) Assigned Friday, October 10th
   (b) Due Monday, October 13th
   (c) Open-book, open-note
   (d) Completed individually
   (e) Submit in PDF form
6. **Magnetostatics, Chapter 5**
   (a) The Lorentz force law
   (b) The Biot-Savart law
   (c) The divergence and curl of magnetic fields
   (d) Magnetic vector potential
   (e) *Homework 4, assigned October 6th*
7. **Magnetic Fields in Matter, Chapter 6**
   (a) Magnetization
   (b) The field of a magnetized object
   (c) The auxiliary field
   (d) Linear and non-linear media
   (e) *Homework 4, due October 17th*
   (f) *Homework 5, assigned October 17th*
8. **Electrodynamics, Chapter 7**
   (a) The electromotive force
   (b) Electromagnetic induction
   (c) Maxwell's equations
   (d) The field of a rotating magnet
   (e) *Homework 5, due October 31st*
   (f) *Homework 6, assigned October 31st*
9. **Computational Electromagnetism Part 1**
   (a) The finite-difference time-domain (FDTD) technique
   (b) Defining boundary conditions in two and three dimensions
   (c) Defining radiating sources
   (d) Examples: point sources and waveguides
   (e) *Homework 6, due November 14th*
10. **Midterm 2, Chapters 5-7**
   (a) Assigned Friday, November 21st
   (b) Due Monday, November 24th
   (c) Open-book, open-note
   (d) Completed individually
   (e) Submit in PDF form
11. **Computational Electromagnetism Part 2**
   (a) Coaxial cables
   (b) Fiber optic couplers
   (c) Radio-frequency (RF) phased arrays
   (d) CEM enhanced with machine learning techniques
   (e) Introduction to CEM research at Whittier College
      i. Research publications
      ii. The Naval Engineering Education Consortium (NEEC) grant
      iii. Educational Partnership Agreement (EPA) internships
   (f) *Homework 7, assigned November 17th*
   (g) *Homework 7, due November 21st*